

CSSE 220 – Object-Oriented Software Development
 Rose-Hulman Institute of Technology

Worksheet 04

Name (Print): _____ Section: _____

1. **Class & Object Terminology**

Term	Notes from lecture
class	
object	
instance	
field	
method	

2. **Class vs Object.** Fill in the blanks.

A(n) _____ is a blueprint.

A(n) _____ is an instance created from the class.

3. **Creating an Object Instance.** Suppose you have a class **Block**. Complete the code to create and use an object.

```
Block dirt = _____; // create object
dirt.type = "Dirt";
```

4. Suppose you have an object called **stone** and the class **Block** defines a method called **mine()**. Now you need to call the mine method.

```
1 Block stone = new Block();
2
3 _____ // // call the mine() method
4
```

5. **Method Templates.** Fill in the missing parts of these Java method templates.

1) This method does not return a value:

```
1 public _____ printBlockInfo() {
2     System.out.println("Block info");
3 }
```

2) This method returns a value (int)

```
1 public _____ getHardness() {
2     return hardness;
3 }
```

6. What is a Constructor?

A constructor is _____

7. Constructor Check. Which of the following is a constructor? (Circle one)

`public void Block() { }`

`public Block() { }`

`public int Block() { return 0; }`

8. Constructor Rules

- The constructor name must match the class name (T / F)
- A constructor has no return type (T / F)
- You always must write a constructor (T / F)
- A Class can have multiple constructors (T / F)

9. The Default (No-Arg) Constructor

If no values are provided, fields receive Java's default values:

- String → _____
- int → _____
- Object → _____

10. Using "this" with a Constructor. Complete the constructor:

```
1 public class Block {  
2     String type;  
3     int hardness;  
4  
5     public Block(String type, int hardness) {  
6  
7         -----  
8  
9         -----  
10    }  
11 }
```

11. Before You Leave

Write one question you still have about classes, objects, or constructors.
